

Impact of Excessive Screen Time on Sleep Quality and Sleep Disturbances Among Young Adults: A Cross-Sectional Study

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ABSTRACT

Background: Excessive screen time, particularly using smartphones, is increasingly recognized as a major contributor to poor sleep quality and daytime sleepiness among young adults. This study investigates the association between smartphone addiction and sleep disturbances in university students.

Methods: A cross-sectional study was conducted among 321 university students aged 18–35 from SRM Medical College Hospital and Research Centre. Participants using digital screens ≥ 3 h/day for the past 3 months were assessed using the Pittsburgh Sleep Quality Index (PSQI), Epworth Sleepiness Scale (ESS), and Smartphone Addiction Scale-Short Version (SAS-SV). Sleep quality, daytime sleepiness, and smartphone addiction levels were analyzed using descriptive statistics, Pearson's correlation, and regression analysis. **Results:** Poor sleep quality (PSQI >5) was reported by 61.7% of participants, while 48.1% exhibited excessive daytime sleepiness (ESS >10). High smartphone usage (SAS-SV >40) was significantly associated with poor sleep quality (mean PSQI = 10.5, $P < 0.01$) and higher daytime sleepiness (mean ESS = 12.1, $P < 0.01$). A moderate positive correlation was observed between SAS-SV and PSQI scores ($r = 0.60$) and a weaker correlation with ESS scores ($r = 0.45$). **Conclusion:** Smartphone addiction is significantly associated with impaired sleep quality and increased daytime sleepiness among young adults.

KEYWORDS: ESS, PSQI, SAS-SV, screen time, sleep quality, smartphone addiction, young adults

INTRODUCTION

Adequate sleep is critical for maintaining both physical and mental health, yet excessive screen time has emerged as a significant disruptor of sleep hygiene among young adults.^[1] The pervasive use of digital devices—particularly smartphones—has been linked to disruptions in melatonin production and circadian rhythms, impairing sleep quality.^[2] Technological advancements have facilitated widespread smartphone adoption, with projections indicating increased usage globally. While smartphones enhance productivity and social connectivity, excessive use can lead to compulsive behaviors categorized as smartphone addiction, which adversely affects sleep patterns.^[3] Evidence suggests that excessive

smartphone usage is increasingly prevalent among students, negatively influencing academic performance and overall well-being.^[4]

This study seeks to explore the connection between smartphone use, sleep disturbances, and overall sleep quality in university students, building on prior research linking excessive screen time to adverse health outcomes.

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Table 1: Sleep quality and daytime sleepiness across levels of smartphone usage (Smartphone Addiction Scale-Short Version categories)

Measure	Low usage (SAS-SV <20, n=32)	Moderate usage (SAS-SV 20–40, n=209)	High usage (SAS-SV >40, n=80)	P
PSQI score, mean (SD)	4.2 (3.1)	7.8 (4.9)	10.5 (5.4)*	<0.01
Poor sleep (PSQI >5), n (%)	12 (37.5)	125 (59.8)	56 (70.0)*	<0.01
ESS score, mean (SD)	7.5 (3.2)	9.9 (4.0)	12.1 (4.3)*	<0.01
Excessive daytime sleepiness (ESS 15–24), n (%)	1 (3.1)	20 (9.6)	12 (15.0)*	<0.05

* $P < 0.05$. SD=Standard deviation, SAS-SV=Smartphone Addiction Scale-Short Version, ESS=Epworth Sleepiness Scale, PSQI=Pittsburgh Sleep Quality Index

Table 2: Prevalence of poor sleep and excessive daytime sleepiness among high smartphone users (Smartphone Addiction Scale-Short Version >40), stratified by gender and field of study

Characteristic	Poor sleep (PSQI >5), n (%)	Excessive daytime sleepiness (ESS 15–24), n (%)	P (PSQI)	P (ESS)
Gender				
Males (n=19)	11 (57.9)	1 (5.3)	<0.05	<0.05
Females (n=61)	45 (73.8)*	11 (18.0)*		
Field of study				
Medical (n=32)	20 (62.5)	3 (9.4)	<0.05	<0.05
Dental (n=35)	26 (74.3)*	7 (20.0)*		

* $P < 0.05$. ESS=Epworth Sleepiness Scale, PSQI=Pittsburgh Sleep Quality Index

METHODS

A cross-sectional study design was employed, involving 321 medical science students—including those from Medicine, Dentistry, Allied Health Sciences, and Paramedical courses—aged 18–26 years. The participants were primarily selected from SRM Medical College Hospital and Research Centre, with the majority representing the medical and dental fields.

The study included young adults aged 18–26 years who used digital screens, such as smartphones, laptops, or tablets, for at least 3 h daily over the preceding 3 months. Participants were required to comprehend and respond to the study questionnaires and provide informed consent.

Sleep quality, daytime sleepiness, and smartphone addiction were evaluated using validated instruments, including the Pittsburgh Sleep Quality Index (PSQI, ranging from 0 to 21, with higher scores indicating poorer sleep quality), the Epworth Sleepiness Scale (ESS, ranging from 0 to 24, with higher scores reflecting greater daytime sleepiness), and the Smartphone Addiction Scale-Short Version (SAS-SV, where higher scores denote a greater risk of smartphone addiction). Excessive screen time was inferred from SAS-SV scores, categorized into three groups: Low Usage (SAS-SV <20), Moderate Usage (SAS-SV 20–40), and High Usage (SAS-SV >40,

indicating excessive screen time and high addiction risk). Sleep quality was classified as Good Sleep (PSQI ≤ 5) or Poor Sleep (PSQI >5), while sleep disturbances were categorized as Normal (ESS 0–10), Moderate (ESS 11–14), or Excessive (ESS 15–24).

RESULTS

Of the 321 participants, the mean screen time (inferred from SAS-SV scores) was approximately 6.4 h/day. A majority (61.7%) reported poor sleep quality (PSQI >5), and 48.1% exhibited excessive daytime sleepiness (ESS >10).

Table 1 illustrates the reveals a significant impact of smartphone usage on sleep quality and daytime sleepiness among 321 university students. High smartphone usage was associated with worse sleep quality, with poor sleep prevalence highest in the high-usage group (70.0%). High usage also correlated with greater daytime sleepiness, with excessive daytime sleepiness more prevalent in high-usage (15.0%) than moderate (9.6%) and low-usage groups (3.1%). The findings suggest a dose-dependent relationship between smartphone usage and sleep disturbances, with high usage significantly exacerbating poor sleep quality and daytime sleepiness.

Table 2 highlights the study reveals that 80 university students with high smartphone addiction have a higher prevalence of poor sleep and excessive daytime sleepiness. Females and dental students are more affected, with a higher prevalence of poor sleep and excessive daytime sleepiness compared to males and medical students. These findings highlight gender and field-specific vulnerabilities to sleep disturbances among high smartphone users, with females and dental students being particularly affected.

Table 3 shows that higher smartphone addiction (SAS-SV scores) is moderately correlated with poorer sleep quality (PSQI; $r = 0.6$, 95% CI: 0.50–0.70), with 70% of high scorers reporting poor sleep. A weaker correlation was found with daytime sleepiness (ESS; $r = 0.45$, 95% CI: 0.35–0.55), with 15% reporting excessive sleepiness. Smartphone addiction appears

Table 3: Estimated associations between smartphone use (Smartphone Addiction Scale-Short Version Score) and sleep outcomes

Relationship	Estimated <i>r</i>	95% confidence interval	Key observation
SAS-SV versus PSQI	0.6	0.50–0.70	70% of participants with high SAS-SV scores reported poor sleep (PSQI >5)
SAS-SV versus ESS	0.45	0.35–0.55	15% of participants with high SAS-SV scores reported excessive daytime sleepiness (ESS >14)

SAS-SV=Smartphone Addiction Scale-Short Version, ESS=Epworth Sleepiness Scale, PSQI=Pittsburgh Sleep Quality Index

more strongly linked to reduced sleep quality than daytime sleepiness.

DISCUSSION

Our study corroborates findings linking excessive screen time, particularly smartphone addiction, to heightened sleep disturbances. The data set analysis revealed that high SAS-SV scores (>40), indicative of excessive screen time, were strongly associated with poor sleep quality (PSQI >5) and increased daytime sleepiness (ESS >14). These findings align with prior research demonstrating that mobile phone overuse significantly correlates with poor sleep quality and daytime sleepiness among students.^[3] The higher prevalence of poor sleep and excessive daytime sleepiness among females and dental students suggests potential gender-specific and field-specific vulnerabilities, possibly due to differences in smartphone usage patterns or academic stress.

The mechanisms driving these associations likely include blue light exposure from screens, which suppresses melatonin production, and compulsive behaviors such as constantly checking social media or prolonged smartphone use before bedtime, as evidenced by responses to SAS-SV Questions 8 and 9. The negative implications of nighttime screen usage are echoed by research indicating that digital exposure during late hours exacerbates insomnia symptoms and poor sleep quality.^[5]

The rise of smartphone addiction among young adults presents a pressing public health concern, as inadequate sleep can exacerbate psychological conditions such as anxiety and depression and impair academic performance.^[6] The stronger correlation between SAS-SV and PSQI ($r = 0.60$) compared with ESS ($r = 0.45$) suggests that screen time more directly impacts sleep quality than daytime sleepiness, which may also be influenced by other factors like academic workload or caffeine use.^[7] These findings highlight the

need for targeted interventions to address smartphone addiction and promote digital hygiene, particularly among high-risk groups like female and dental students.

CONCLUSION

In summary, excessive screen time, driven by smartphone addiction, is closely linked to poor sleep quality and increased daytime sleepiness among young adults.

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Conflicts of interest

There are no conflicts of interest.

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